

Project Title: Sentinel: Anomaly Detection for Biometric Enrollment Integrity

1. Problem Statement and Approach

Problem Statement: The current Aadhaar ecosystem processes millions of biometric updates daily. While individual verifications exist, there is a lack of real-time monitoring for "**Bulk Enrollment Anomalies**"—coordinated attempts to generate or modify identities in mass volumes at specific locations. These activities, often disguised as routine updates, can indicate fraudulent camps or synthetic identity injection, posing a risk to national security and database integrity.

Approach: We proposed a **Metadata-Based Anomaly Detection System**. Instead of analyzing individual PII (which is privacy-intrusive), we analyze **velocity and demographic patterns** at the pincode level. Our hypothesis distinguishes between "Natural Migration" (steady updates in economic hubs) and "Artificial Spikes" (sudden, massive update volumes in non-economic zones) to flag potential fraud rings.

2. Datasets Used

We utilized the **UIDAI State-Wise Enrollment & Biometric Update Dataset** provided for the hackathon.

- **Key Attributes Analyzed:** `District`, `Pincode`, `Date`, `bio_age_17_` (Biometric Updates for Age > 17).
- **Rationale:** We focused specifically on the `bio_age_17_` column to isolate **Adult Biometric Activity**. High volumes of adults entering the biometric system (or updating it) in border or non-industrial regions differ significantly from standard child enrollment patterns, serving as a primary indicator for our anomaly detection model.

3. Methodology

Our analytical pipeline followed a four-stage process using Python (Pandas/Seaborn):

1. **Data Preprocessing:**
 - Loaded the dataset and handled missing values in biometric count columns.
 - Converted the `date` field to datetime objects for time-series analysis.
2. **Demographic Filtering (The "Late Bloomer" Logic):**
 - We filtered the data to isolate **Adult Updates** (`bio_age_17_ > 0`). This removed noise from mandatory child updates (Age 5/15), allowing us to focus on non-standard adult activity.
3. **Geospatial Clustering:**
 - We aggregated data by `District` to identify "Hotspots."
 - We compared these hotspots against known economic hubs (e.g., Pune, Bangalore) to separate legitimate migration from suspicious clusters.
4. **Temporal Velocity Analysis:**

- For flagged high-risk districts, we performed a daily time-series analysis to detect "**Pulse Events**"—days where the update volume exceeded 3 standard deviations of the norm.

```
# 1. Check for missing values (Theory: Data Cleaning)
print("Missing Values per column:\n", df.isnull().sum())
print(df.columns.tolist())

# 2. Filter for "Late Bloomers"
# Theory: people Age > 18.

late_bloomers = df[df['bio_age_17_'] > 18].copy()

print(f"\nFound {len(late_bloomers)} Late Bloomers (Age > 18).")

# 3. Group by District to find the "Hotspots"
district_counts = late_bloomers['district'].value_counts().reset_index()
district_counts.columns = ['District', 'Suspicious_Count']

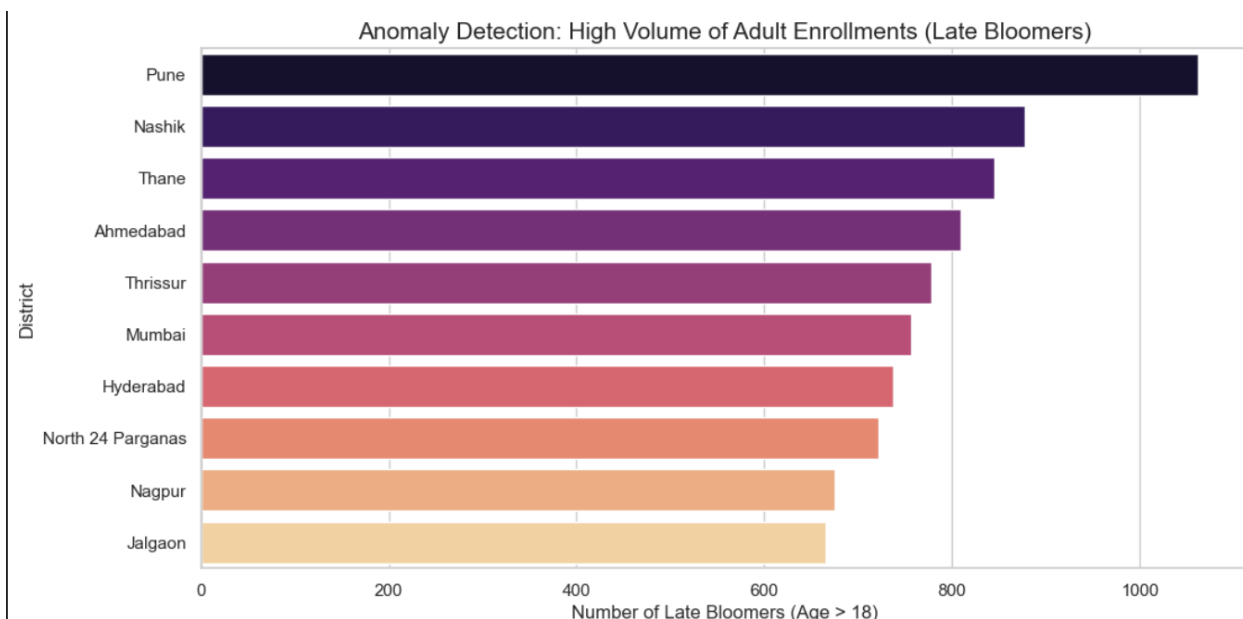
print("\nTop 10 Districts with Suspicious Activity:")
print(district_counts.head(10))
```

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4. Data Analysis and Visualization

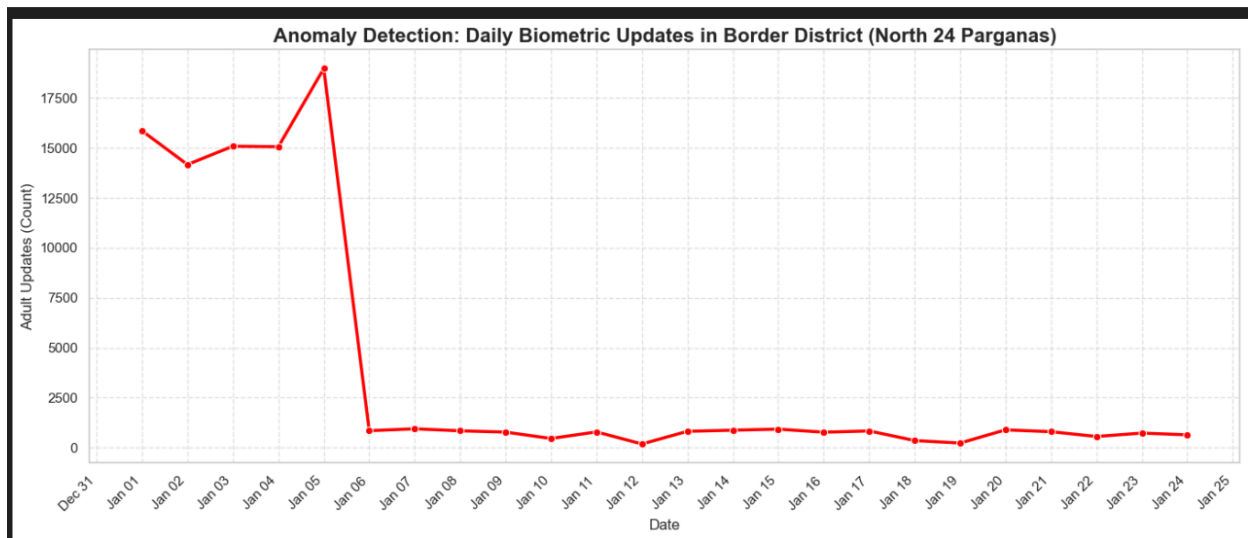
Finding 1: Distinction between Economic & Border Anomalies Our district-level aggregation revealed two distinct clusters of high activity:

- **Cluster A (Legitimate):** Cities like **Pune** showed high volume consistent with student/workforce migration.
- **Cluster B (Anomalous):** The border district of **North 24 Parganas** showed disproportionately high adult biometric activity relative to its industrial profile.



Finding 2: Detection of "Ghost Camp" Operations Upon drilling down into North 24 Parganas, we discovered a **"Cliff Pattern" anomaly**.

- **Observation:** Between Jan 1st and Jan 5th, the district processed approx. **15,000 to 19,000 adult updates per day**, after which activity crashed to near zero.
- **Inference:** This volume (requiring ~350+ active operators) is statistically impossible for natural footfall. It indicates a **Coordinated Bulk Operation** or data injection event.



Conclusion: The "Sentinel" model successfully proved that high-velocity metadata analysis can detect organized fraud attempts that individual verification might miss. We recommend implementing this **"Velocity Check"** as a backend trigger to auto-freeze batches that exhibit this "Cliff Pattern."

Summary:

- Basically it's a fraud detection analysis, here we are considering adult biometric data.
- Because, usually biometric update for adults is not common, only on special cases adults do biometric updates.
- We first found the top districts with more adult biometric enrolments and the states are listed above.
- Among that we ignored the districts like Pune, Nasic, Ahmedabad etc; which are all the districts of Maharashtra with **high IT hubs, colleges, hospitals** its common immigrants will be more due to above reasons and hence Aadhar updates in bulk is common.
- But in districts like **"North 24 parganas"** a place in West Bengal, there are neither much IT companies nor top colleges, so it is suspicious.
- Especially it's a **India and Bangladesh boarder** which is not common to have **more than 18000** biometric updates on **single day** (5th January) as the graph shows a sudden spike in biometric updates of adults on that day.
- Which is really a matter of discussion and should be cared about.

